

Supplementary Material

Viewpoint Invariance Is a Theorem, Not a Fit

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This supplement carries the depth condensed out of the main paper: the proof of the invariance theorem (Sec. S1), the full invariant-set specification (Sec. S2), the eight-gate numerical certificate (Sec. S3), the complete precision budget (Sec. S4), the nondeterminism seed distribution (Sec. S5), the protocol audit and continuous-time control (Sec. S6), the units audit behind our comparison against published KIMORE numbers (Sec. S7), the per-family ablation of the invariant cut (Sec. S8), the in-dataset score-resolution analysis behind the clinical-magnitude claim (Sec. S9), and the full second-corpus node-failure curve (Sec. S10). All numbers are reproduced from the same banked artifacts as the main paper; equation and proposition references (e.g. Prop. 1) point to the main text.

S1. Proof of the Invariance Theorem

Proposition 1 ($SO(3)$ -invariance of the read-out; = Prop. 1). *Let the encoder intertwine with the Wigner-D action, so under $g \in SO(3)$ each latent transforms as $s_j \mapsto s_j$ (type-0) and $v_j \mapsto D_1(g) v_j$ (type-1), with $D_1(g) \in SO(3)$ orthogonal and $\det D_1(g) = +1$. Then the projection Π (main Eq. 1–2) is invariant under $SO(3)$ for arbitrary encoder weights, and the predicted score is invariant to any rigid camera motion.*

Proof. Feeding root-relative coordinates makes every input a difference $\tilde{x}_i - \tilde{x}_j$, so a camera translation cancels; it remains to treat rotations. Consider each generator of Eq. 1 under $v \mapsto D_1(g)v$ with $R := D_1(g)$ orthogonal: (i) $\tanh s_j$ is a function of a type-0 scalar, hence unchanged. (ii) $\|v_j^c\| = \|Rv_j^c\|$ since $R^\top R = I$; so $\log(1 + \|v\|)$ is invariant. (iii) $\langle \widehat{Rv}, \widehat{Rv'} \rangle = \langle Rv/\|v\|, Rv'/\|v'\| \rangle = \langle \hat{v}, \hat{v'} \rangle$, again because R preserves inner products. (iv) $\det[\widehat{Rv^a}, \widehat{Rv^b}, \widehat{Rv^c}] = \det(R) \det[\hat{v}^a, \hat{v}^b, \hat{v}^c] = \det[\hat{v}^a, \hat{v}^b, \hat{v}^c]$, using $\det(R) = +1$; the triple products are invariant under *proper* rotations (and flip sign under a reflection, $\det = -1$, which is exactly why they narrow the guaranteed group from $O(3)$ to $SO(3)$). The bone lengths $d_{ij} = \|\tilde{x}_i - \tilde{x}_j\|$ and the

Table S1. Structural specification of the invariant cut Π .

Generator	Irrep	Count	Dim
<i>Parity-even</i> — generating set complete for $O(3)$			
scalars $\tanh s$	$0e$	n_0	32
vector norms $\log(1 + \ v\)$	$0e$	n_1	8
cosines $\langle \hat{v}, \hat{v'} \rangle$	$0e$	$\binom{n_1}{2}$	28
<i>pooled</i> mean $\oplus$ max		2×68	136
bone lengths d_{ij}	$0e$	per bone	24
even subtotal			160
<i>Parity-odd</i> — $SO(3)$ chiral completion			
triple products $\det[\hat{v}, \hat{v'}, \hat{v'']}$	$0o$	$\binom{n_1}{3}$	56
<i>pooled</i> mean $\oplus$ max		2×56	112
anatomical volumes χ_τ	$0o$	per frame	11
odd subtotal			123
full $SO(3)$ cut			283

anatomical signed volumes χ_τ (normalised determinants of joint triads) are invariant by the same orthogonality and $\det(R) = +1$ arguments. Finally mean_j and max_j are applied identically to invariant per-joint quantities, hence themselves invariant. Every entry of Π is therefore fixed by g , for any weights; the head is a deterministic function of Π , so the score is invariant. $\square$ $\square$

S2. Full Invariant-Set Specification

Table S1 states both generating sets explicitly at the deployed widths ($n_0=32$, $n_1=8$; $J=25$ joints, 24 bones, $|\mathcal{T}|=11$ anatomical frames). Both parity classes share one mean $\oplus$ max joint pool; bone lengths and signed volumes are concatenated index-preserved (unpooled). The parity-even subtotal (160) is exactly the projection whose precision budget is audited in Sec. S4; the parity-odd completion adds 123 dimensions.

S3. Numerical Certificate (E1–E8)

We certify the *deployed* model. E1–E5 establish that the claimed invariance is present; E6–E8 establish that it is the

- one we actually have. The three conditions of E3–E4 — violation at roundoff, *flat* in angle, and $r \gg 1$ — are what upgrade “small” to “zero”: magnitude alone would not. Reported figures are from a representative fp64 run; being roundoff-floor quantities they are *regime*-stable (order of magnitude, sign of slope, $r \gg 10^5$) but not bit-reproducible.
- E1 **Representation orthogonality:** $\max_g \|\rho(g)^\top \rho(g) - I\|_F \leq 10^{-12}$ (fp64).
- E2 **Invariance of the motion channels:** δ, σ verified invariant, not assumed.
- E3 **Rotation sweep:** score violation $\sim 10^{-16}$ across $\theta \in [0.05, 3.0]$ rad, with log–log slope $|\text{slope}| < 0.5$ (flat). A modelling violation grows with angle; roundoff does not.
- E4 **Precision scaling:** $r = \bar{\delta}^{\text{fp32}} / \bar{\delta}^{\text{fp64}} = 5.44 \times 10^{-8} / 4.43 \times 10^{-16} = 1.23 \times 10^8$, tracking the machine-epsilon ratio; a genuine violation would be precision-independent ($r \approx 1$).
- E5 **Translation invariance:** on raw world coordinates, $|s(\mathbf{x} + \mathbf{c}) - s(\mathbf{x})| / |s(\mathbf{x})| = 2.6 \times 10^{-16}$.
- E6 **Parity:** under a reflection the parity-even cut gives *exactly* 0 (bit-identical on a mirrored body — an unintended $O(3)$ -invariance), whereas the completed cut gives 4.4×10^{-2} (seed 0, `certify_egru.json`). A single reflection certifies the entire improper coset.
- E7 **Laterality:** permuting left/right joint *indices* (a relabelling, not a reflection) moves the score by 6.9×10^{-2} / 5.6×10^{-2} for the two cuts (seed 0); both separate a left- from a right-limb movement. E6–E7 are model-dependent modelling quantities, not roundoff-floor ones, and vary across seeds.
- E8 **The fix is free:** read-out invariance under Haar-random *proper* rotations (up to $\theta = \pi$), with the parity-odd channels active, is 2.9×10^{-16} — unchanged. E1–E5 are likewise over Haar-random $SO(3)$, so the certified invariance is to the full rotation group, of which a camera azimuth or the NTU cross-view displacement is one instance. We report no separate NTU-specific invariance measurement: the gates are architectural and the corpus does not change them.

S4. Complete Precision Budget

The full fp64→int8 budget — including the bf16 and int8-activations-only rows — is reported in the main paper’s deployment table. The int8 cliff is attributable *solely* to activation grid-rounding: weight-only quantization preserves the theorem (a quantized model is a different equivariant model), and fp16 is $3.2\times$ tighter than bf16 because equivariance pays in mantissa bits, which bf16 trades for exponent range.

Table S2. Per-(seed,fold) clean MAD, $SO(3)$ EGRU. No retraining beyond the three seeds.

fold	seed 0	seed 1	seed 2	std	gap
0	7.120	6.778	6.737	0.210	0.383
1	6.680	6.124	5.804	0.443	0.876
2	6.976	7.345	6.623	0.361	0.722
3	6.762	7.363	6.093	0.635	1.270
4	6.469	7.714	6.415	0.735	1.299
mean across-seed std				0.48	

S5. Nondeterminism: Seed Distribution

The 0.33 MAD floor quoted in the main paper is the tighter *fixed-configuration* run-to-run component (init held constant, atomic nondeterminism only). Table S2 gives the wider *seed-to-seed* distribution ($SO(3)$ EGRU clean MAD per fold across three initializations): mean across-seed std 0.48, worst pairwise gap 1.30. Both exceed every model-vs-model clean gap in the main accuracy table, so those gaps are ties. cuDNN-deterministic modes remove the cuDNN-GRU atomic component but not the initialization variance; the structural degradation columns are weight-independent (Prop. 1 holds for arbitrary weights), so the flat viewpoint curve stayed exactly flat across every seed while absolute MAD moved.

S6. Protocol Audit and Continuous-Time Control

Protocol audit. The single-exercise audit of the reference point-cloud transformer — selection bias $+1.2 \pm 0.4$ MAD (18.6% over three seeds), paired subject-cluster bootstrap $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$, clearing zero in only one of three seeds) — is reported per-seed in the main paper. It is *not* subject leakage (each subject contributes one recording on that slice, so a random split is already subject-disjoint); the slice simply carries no robust subject-level signal, so we pool the five exercises and gate every experiment on a per-exercise mean-predictor floor. Pooled, every model clears that floor with its bootstrap interval entirely below zero, so the fragility is a property of the slice and not of any architecture evaluated on it.

Continuous-time control. We implement the Neural CDE variant and certify it to the same E1–E8 standard; it satisfies an exact equivariance certificate and *fails* its own measured floor (8.43 vs. 8.25 MAD), the cleanest demonstration that a structural guarantee is worthless without measured signal. Analysing the adaptive solver, a Dormand–Prince controller accepts/rejects steps on a per-component scaled-RMS norm that is *not* group-invariant even when the vector field is exactly equivariant, so rotated and unrotated integrations receive different step grids and de-align;

a per-irrep-normalised norm removes this, driving grid divergence to identically zero.

Irregular-sampling null (derived). A model-free spectral census finds 97.7% of KIMORE’s *positional* energy below the fixed-grid resampling corner (though 70% of *velocity* energy lies above it), so resampling preserves pose geometry while low-passing the derivative band and acts as a denoiser on healthy-form exercises, where the discriminative signal is positional — the null was predictable before the experiment, though the boundary would move for tremor-dominated tasks. The mechanism is nonetheless real: a physiologically calibrated 5 Hz tremor is recovered from the true irregular stamps at $r=1.000$ under drop rates up to 70%, while a resampled estimator decays to 0.785.

S7. Units of the Published KIMORE Numbers

The main text compares our MAD against published KIMORE MAD directly. That is only legitimate if both are in clinical-score units, and none of those papers states the scale. The question is not academic: the reference point-cloud transformer standardises its targets with a `StandardScaler` ($\sigma = 8.466$ on `Kimore_ex1`) and its per-epoch training print is in standardised units while its evaluator inverse-transforms, so the two readings of the same headline number differ by $8.5\times$.

The published tables settle it without recourse to anyone’s source code. Every method in this lineage reports MAD and MAPE for the same run, with $\text{MAPE} = 100 \cdot \text{mean}(|y - \hat{y}|/|y|)$, so

$$|\hat{y}| = 100 \cdot \text{MAD}/\text{MAPE} \quad (1)$$

recovers the magnitude of whatever array was handed to the metric. On raw clinical scores it must land near KIMORE’s per-exercise means; on standardised targets it must land near $\text{mean}|z| \approx 0.8$.

Table S3 is unambiguous: the median recovered magnitude over all 40 published cells is 38.1 against a true per-exercise mean of 35.0–40.7 (median relative error 10.6%), while the standardised reading is excluded by at least $70\times$ — it would require these papers to have printed MAPE values in the tens of percent rather than the 0.17–6.0 they report. The published KIMORE MAD numbers are therefore in 0–50 clinical-score units, and our 6.3–6.7 is on the same axis.

The convention is also documented at its source, so the inference above is corroborated rather than load-bearing. Both papers load KIMORE through descendants of a single `DataLoader`, the one released with the STGCN-rehab benchmark (Deb et al., IEEE TNSRE 2022); the point-cloud transformer’s copy is near-identical line for line. That release’s training script inverse-transforms predictions *and* targets before computing MAD, RMSE and

$ \hat{y} = 100 \text{ MAD}/\text{MAPE}$	Ex1	Ex2	Ex3	Ex4	Ex5
Du et al. 2021	39.4	36.6	32.8	34.1	33.2
Deb et al. 2022	41.5	60.8	51.4	42.1	39.0
Yao et al. 2023	40.2	35.1	32.5	35.5	36.1
Mourchid et al. 2023	39.5	77.3	34.3	38.1	32.8
Zhang et al. 2025	41.2	51.6	38.4	42.2	35.0
Kuang et al. 2025	43.2	31.4	41.0	30.6	32.2
Kuang et al. 2026	54.5	40.1	48.3	63.4	18.8
Point-cloud transf.	34.1	29.6	38.1	33.4	32.4
true score mean	40.7	35.6	39.5	35.0	36.4
true mean $ z $	0.80	0.87	0.79	0.86	0.84

Table S3. Target magnitude recovered from each paper’s own published MAD/MAPE pair, against KIMORE’s true clinical Total Score statistics. Every row lands in the score band (30–63), none anywhere near the standardised band (≈ 0.8). Rows reporting no MAPE are omitted.

MAPE, and its released label file holds raw clinical Total Scores. The reporting convention is therefore fixed upstream of any individual paper, and Tab. S3 confirms it independently, from published numbers alone. The audit is reproduced by `src/published_units_audit.py`, which gates on all three conditions.

S8. Per-Family Ablation of the Invariant Cut

Each family of the invariant cut (Sec. S2) is zeroed at inference on the deployed chiral model, dimension preserved and with no retraining, resolving the mechanism to family granularity. The main paper summarises the result; Tab. S4 is the full table.

Family removed	Parity	Clean	ΔAcc	nf lost
— (full cut)	—	6.73	—	+3.76
cosines	even	8.56	+1.83	+7.29
triple products	odd	9.48	+2.75	−0.16
bone lengths	even	7.40	+0.67	+2.36
scalars	even	6.98	+0.25	+3.46
vector norms	even	6.89	+0.16	+3.51
anat. volumes	odd	6.93	+0.19	+3.43

Table S4. Per-family ablation of the cut (each family zeroed at inference; 3 seeds $\times$ 5 folds). ΔAcc below the 0.33 floor is a tie; “nf lost” is worst MAD over 8 dead joints (full cut +3.76).

S9. Score Resolution, and Why Not Clinical Bands

The natural way to give the MAD scale an in-dataset meaning is to ask whether a score swing moves a patient across KIMORE’s clinical bands. We do not do this, because KIMORE has no such bands. The five group labels record how

a subject was *recruited*, not what score range they occupy, and on every one of the five exercises the lowest-scoring control sits below the highest-scoring patient (Tab. S5). Assigning bands to these groups would require inventing the boundaries, which is the same fabrication we decline elsewhere in this paper.

	lowest control	highest patient	margin
Es1	34.00 (NE_ID10)	50.00 (P_ID3)	+16.00
Es2	23.00 (NE_ID10)	50.00 (B_ID7)	+27.00
Es3	30.00 (NE_ID10)	48.00 (P_ID1)	+18.00
Es4	26.00 (NE_ID11)	42.33 (B_ID6)	+16.33
Es5	19.67 (NE_ID2)	50.00 (P_ID6)	+30.33

Table S5. Clinical Total Score, control vs. patient, per exercise. Every row is inverted, by 16 to 30 points of fifty. The group labels are not score strata.

We therefore measure the swing against the score’s *resolution*: how large it is relative to the differences the score is being asked to express. The between-subject sd of the Total Score is 10.01 (mean over the five exercises; 10.26 pooled), so the augmented baseline’s mean per-sequence swing of 3.01 is 0.30 sd, its 95th percentile 8.65 is 0.86 sd, and its worst sequence 20.12 is 2.01 sd. The ordinal form of the same statement is Tab. S6: a pair of subjects separated by less than the swing is a pair whose ordering a camera relocation can invert, and ordinal comparison — is this patient better than that one, or than they were last week — is what these scores are used for.

% of subject pairs closer than	<i>n</i> pairs	3.01	8.65	20.12
all pairs	14251	20.2	47.7	83.0
cross-group only	11014	17.4	43.1	80.4
patient vs. control	7122	12.8	34.3	74.2

Table S6. Fraction of within-exercise subject pairs whose Total Scores are separated by less than the augmented baseline’s mean (3.01), 95th-percentile (8.65) and worst-case (20.12) per-sequence viewpoint swing. These are the pairs the camera can reorder. Pairs are formed within an exercise, since scores are comparable only across subjects performing the same task.

A third, same-patient comparator avoids between-subject variation entirely. Each subject is rated on five exercises, and the median spread of a subject’s own five scores is 4.97 sd (12.00 range). The swing’s 95th percentile is 1.74× that median within-subject sd and its worst case 4.05×, so at the tail a camera relocation perturbs the score by several times the variation a single patient already exhibits. This is a conservative yardstick: different exercises are different measurements, so the within-subject spread is an upper bound on within-subject noise, which makes the ratio a lower bound on the swing’s severity.

Scope and census. This measures score *resolution*, not group *discriminability* — the claim is that a camera move perturbs the clinical output by an amount comparable to the differences that output expresses, not that the score classifies pathology, which Tab. S5 shows it does not. The analysis uses the 380 subject–exercise cells present in the released data (77 subjects; 5 of the nominal 385 cells are absent upstream). Five subjects (B_ID3, B_ID4, P_ID10, S_ID5, S_ID6) carry scores that are not multiples of 1/3, unlike every other subject’s rater averages; excluding them moves nothing material (mean swing 0.30 sd, 21.3% of pairs reorderable at 3.01), and both readings are emitted by `src/clinical_resolution.py`. The swing values are read from the banked Block-3 aggregate rather than recomputed, so this analysis cannot drift from the numbers quoted in the main text.

S10. Node-Failure Curve and Which Spread We Quote

The main paper summarises second-corpus node failure with one number, AUROC lost from $k=0$ to $k=8$ frozen joints. That scalar hides the shape, and the shape is not uniformly in our favour.

REHAB24-6, AUROC at k frozen joints	$k=0$	$k=1$	$k=2$	$k=4$	$k=8$
EGRU (ours)	0.738	0.714	0.694	0.659	0.606
PCT (baseline)	0.707	0.697	0.685	0.660	0.606

Table S7. The full curve behind the single “AUROC lost” figure. Mean over 3 seeds × 5 subject-disjoint folds.

Table S7 shows the two models *cross* between $k=2$ and $k=4$: we start 0.031 ahead, are level at $k=4$ (0.659 vs. 0.660), and are 0.021 behind at $k=8$. This is the same Pareto picture the main paper states for KIMORE — the point-cloud transformer is the more node-robust model, and our claim is a trade, not a dominance. Reporting only the endpoint difference would have made the trade look like a loss on one axis rather than a crossover with a locatable knee.

Which spread the $\pm$ denotes. These runs admit two variances that differ by roughly 5×, and they answer different questions. The *fold-level* sd — the spread of 5 subject-disjoint folds within one seed — is 0.060 (EGRU) and 0.065 (PCT); a fold holds out 2 of 10 subjects, so this is large by construction. The *seed-level* sd — the spread of the 3 per-seed means — is 0.010 and 0.023, because averaging five folds cancels most of the subject variation. The $\pm$ quoted in the main paper is the seed-level one: it describes *seed reproducibility*, not confidence in the AUROC given this subject pool. At 10 subjects the fold-level figure is the honest guide to how far these numbers would move on a different cohort, which is why this corpus is presented as a structural

286 replication and not as an accuracy benchmark. Both are re-
287 generated by `src/second_corpus_tables.py`.